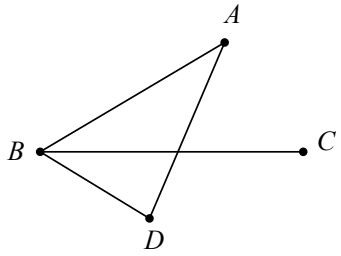
Power Question 2012: Cops and Robbers

This Power Question involves one Robber and one or more Cops. After robbing a bank, the Robber retreats to a network of hideouts, represented by dots in the diagram below. Every day, the Robber stays holed up in a single hideout, and every night, the Robber moves to an adjacent hideout. Two hideouts are *adjacent* if and only if they are connected by an edge in the diagram, also called a *hideout map* (or *map*). For the purposes of this Power Question, the map must be connected; that is, given any two hideouts, there must be a path from one to the other. To clarify, the Robber may not stay in the same hideout for two consecutive days, although he may return to a hideout he has previously visited. For example, in the map below, if the Robber holes up in hideout C for day 1, then he would have to move to B for day 2, and would then have to move to either A, C, or D on day 3.

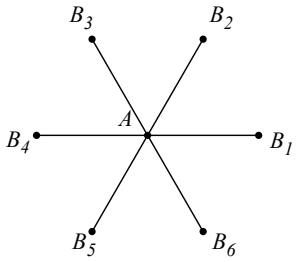


Every day, each Cop searches one hideout: the Cops know the location of all hideouts and which hideouts are adjacent to which. Cops are thorough searchers, so if the Robber is present in the hideout searched, he is found and arrested. If the Robber is not present in the hideout searched, his location is *not* revealed. That is, the Cops only know that the Robber was not caught at any of the hideouts searched; they get no specific information (other than what they can derive by logic) about what hideout he was in. Cops are not constrained by edges on the map: a Cop may search any hideout on any day, regardless of whether it is adjacent to the hideout searched the previous day. A Cop may search the same hideout on consecutive days, and multiple Cops may search different hideouts on the same day. In the map above, a Cop could search A on day 1 and day 2, and then search C on day 3.

The focus of this Power Question is to determine, given a hideout map and a fixed number of Cops, whether the Cops can be sure of catching the Robber within some time limit.

Map Notation: The following notation may be useful when writing your solutions. For a map M, let $h(M)$ be the number of hideouts and $e(M)$ be the number of edges in M. The *safety* of a hideout H is the number of hideouts adjacent to H, and is denoted by $s(H)$.

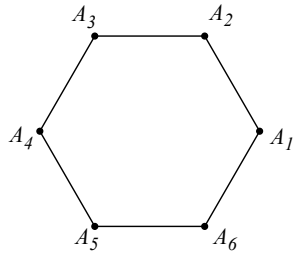
1a. Consider the hideout map M below.



Show that one Cop can always catch the Robber. [1]

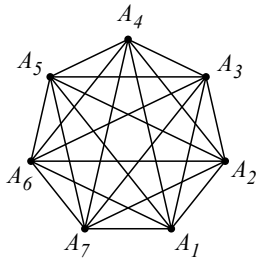
The *Cop number* of a map M, denoted $C(M)$, is the minimum number of Cops required to guarantee that the Robber is caught. In 1a, you have shown that $C(M) = 1$.

1b. The map shown below is $\mathcal{C}_6$, the cyclic graph with six hideouts. Show that three Cops are sufficient to catch the Robber on $\mathcal{C}_6$, so that $C(\mathcal{C}_6) \leq 3$. [2]

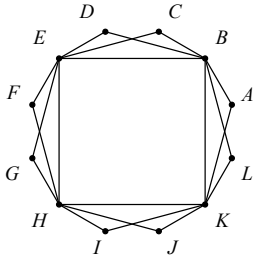


1c. Show that for all maps M, $C(M) < h(M)$. [2]

2a. Find $C(M)$ for the map below. [2]



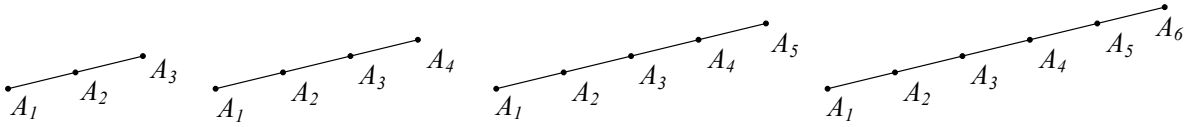
2b. Show that $C(M) \leq 3$ for the map below. [2]



2c. Generalize the result of 2a: if $\mathcal{K}_n$ is a map with n hideouts in which every hideout is adjacent to every other hideout (also called the *complete map* on n hideouts), determine $C(\mathcal{K}_n)$. [3]

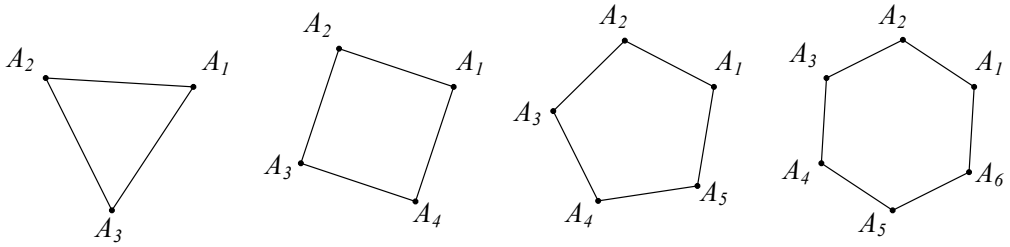
The police want to catch the Robber with a minimum number of Cops, but time is of the essence. For a map M and a fixed number of Cops $c \geq C(M)$, define the *capture time*, denoted $D(M, c)$, to be the minimum number of days required to *guarantee* a capture using c Cops. For example, in the graph from 1b, if three Cops are deployed, they might catch the Robber in the first day, but if they don't, there is a strategy that will guarantee they will capture the Robber within two days. Therefore the capture time is $D(\mathcal{C}_6, 3) = 2$.

3. A *path on n hideouts* is a map with n hideouts, connected in one long string. (More formally, a map is a path if and only if two hideouts are adjacent to exactly one hideout each and all other hideouts are adjacent to exactly two hideouts each.) It is denoted by $\mathcal{P}_n$. The maps $\mathcal{P}_3$ through $\mathcal{P}_6$ are shown below.



Show that $D(\mathcal{P}_n, 1) \leq 2n$ for $n \geq 3$. [4]

4. A *cycle on n hideouts* is a map with n hideouts, connected in one loop. (More formally, a map is a cycle if every hideout is adjacent to exactly two hideouts.) It is denoted by $\mathcal{C}_n$. The maps $\mathcal{C}_3$ through $\mathcal{C}_6$ are shown below.



a. Determine $C(\mathcal{C}_n)$ for $n \geq 3$. (Note that this is *determine* and not *compute*.) [2]

b. Show that $D(\mathcal{C}_n, C(\mathcal{C}_n)) < \frac{3n}{2}$ for $n \geq 3$. [2]

Definition: The *workday number* of M, denoted $W(M)$, is the minimum number of *Cop workdays* needed to guarantee the Robber's capture. For example, a strategy that guarantees capture within three days using 17 Cops on the first day, 11 Cops on the second day, and only 6 Cops on the third day would require a total of $17 + 11 + 6 = 34$ Cop workdays.

5. Determine $W(M)$ for each of the maps in problem 2a and 2b. [4]

6. Let M be a map with $n \geq 3$ hideouts. Prove that $2 \leq W(M) \leq n$, and that these bounds cannot be improved. In other words, prove that for each $n \geq 3$, there exist maps M_1 and M_2 such that $W(M_1) = 2$ and $W(M_2) = n$. [6]

Definition: A map is *bipartite* if it can be partitioned into two sets of hideouts, $\mathcal{A}$ and $\mathcal{B}$, such that $\mathcal{A} \cap \mathcal{B} = \emptyset$, and each hideout in $\mathcal{A}$ is adjacent only to hideouts in $\mathcal{B}$, and each hideout in $\mathcal{B}$ is adjacent only to hideouts in $\mathcal{A}$.

7a. Prove that if M is bipartite, then $C(M) \leq n/2$. [3]

7b. Prove that $C(M) \leq n/2$ for *any* map M with the property that, for all hideouts H_1 and H_2, either all paths from H_1 to H_2 contain an odd number of edges, or all paths from H_1 to H_2 contain an even number of edges. [3]

8. A map M is called *k-perfect* if its hideout set H can be partitioned into equal-sized subsets $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_k$ such that for any j, the hideouts of $\mathcal{A}_j$ are only adjacent to hideouts in $\mathcal{A}_{j+1}$ or $\mathcal{A}_{j-1}$. (The indices are taken modulo k: the hideouts of $\mathcal{A}_1$ may be adjacent to the hideouts of $\mathcal{A}_k$.) Show that if M is k-perfect, then $C(M) \leq \frac{2n}{k}$. [4]

9. Find an example of a map M with 2012 hideouts such that $C(M) = 17$ and $W(M) = 34$, or prove that no such map exists. [5]

10. Find an example of a map M with 4 or more hideouts such that $W(M) = 3$, or prove that no such map exists. [5]